TUTORIAL 7

Rank, Row and Column Spaces, and Measurement Systems

MOTIVATION AND GUIDING IDEA

Motivation. In $Ax = b$, the column space determines attainable measurements, while the kernel determines undetectable changes in the state. Distinguishing these roles allows existence and uniqueness to be tested before solving explicitly.

NOTATION AND CONVENTIONS

- $\mathbb{F}^{m\times n}$ is the space of matrices with m rows and n columns over the field $\mathbb{F}$; A^T denotes the transpose.
- $\mathcal{R}(A) = \text{Row}(A) \subseteq \mathbb{F}^n$ is the row space; $\mathcal{C}(A) = \text{Col}(A) \subseteq \mathbb{F}^m$ is the column space. Identify a row with its transposed column when matrix multiplication requires it.
- $N(A) = \ker A = \{x \in \mathbb{F}^n : Ax = 0\}$ is the kernel; $\text{nullity}(A) = \dim \ker A$ is the nullity; $\text{rank}(A) = \dim \text{Col}(A) = \dim \text{Row}(A)$ is the rank.
- span S consists of all finite linear combinations of vectors in S; e_j is the standard basis vector with entry 1 at index j and 0 elsewhere.
- $x_0 + \ker A = \{x_0 + v : v \in \ker A\}$ is an affine space, a translate of the kernel. Its affine dimension is $\dim \ker A$; it need not be a vector subspace.

DEFINITION 7.1

Let $A \in \mathbb{F}^{m\times n}$, and define $T_A : \mathbb{F}^n \to \mathbb{F}^m$ by $T_A(x) = Ax$. Then $\text{Im}\, T_A = \text{Col}(A)$ and $\ker T_A = N(A)$, so

$$\text{rank}(A) + \dim \ker A = n.$$

The right-hand side is the dimension of the **domain**. The system $Ax = b$ is solvable if and only if $b \in \text{Col}(A)$. If x_0 is one solution, **all** solutions form $x_0 + \ker A$. An existing solution is therefore unique if and only if $\ker A = \{0\}$. The nonzero echelon rows form a basis of the row space, but a column-space basis must use the **original** columns at the pivot indices.

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MOTIVATION AND GUIDING IDEA

Tutorial objectives. Distinguish the row space, column space and null space, and use rank and nullity to quantify information and freedom in a system. The measurement examples apply to engineering systems in many disciplines.

7.1 Row and column spaces and the rank theorem

RECALL AND TOOLS

The **row space** $\mathcal{R}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^n$ spanned by its rows. .

- Elementary row operations preserve the row space of a matrix.

PROPOSITION 7.1

- **Claim 1.** If $A, B \in M_{m\times n}(\mathbb{F})$ are row equivalent, then $\mathcal{R}(A) = \mathcal{R}(B)$.
- **Claim 2:** If A is in row echelon form, its nonzero rows form a basis of $\mathcal{R}(A)$, (and are therefore linearly independent in $\mathbb{F}^n$).

RECALL AND TOOLS

The **column space** $\mathcal{C}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^m$ spanned by its columns. .

- Row reduction **does not in general preserve the column space**; only its **dimension** is preserved.
- $\mathcal{C}(A) = \mathcal{R}(A^T)$ when a row vector is identified with its transposed column vector.

PROPOSITION 7.2

The rank theorem: Every matrix A satisfies $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$. This common dimension is called the **rank of the matrix**.

RECALL AND TOOLS

- The row and column spaces need not be the same subspace. For a rectangular matrix they even lie in different ambient spaces, $\mathbb{F}^n$ and $\mathbb{F}^m$, respectively; their dimensions are nevertheless equal.
- Equivalently, rank$A$ is the number of pivots in the reduced row echelon form.
- For $A \in \mathbb{R}^{m\times n}$, $\text{rank}(A) \leq \min(m, n)$.
- A square matrix $A \in \mathbb{R}^{n\times n}$ is invertible if and only if $rank(A) = n$
- $rank(A \cdot B) \leq min\{rank(A), rank(B)\}$
- $rank(A + B) \leq rank(A) + rank(B)$

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EXERCISE 7.1

Please find a basis and the dimension of the row space and column space of

$$A = \begin{pmatrix} 3 & -6 & 9 \\ 1 & -2 & 3 \end{pmatrix}.$$

WORKED SOLUTION

Interchange the rows, then subtract three times the first row from the second:

$$\begin{pmatrix} 3 & -6 & 9 \\ 1 & -2 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 3 \\ 3 & -6 & 9 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Row operations preserve the row space, so $\{(1, -2, 3)\}$ is a basis for it. For the column space, choose the original column at the pivot index: $\{(3, 1)^T\}$. Indeed, the second and third columns are -2 and 3 times the first. Both spaces have dimension 1, but the rows lie in $\mathbb{R}^3$ and the columns in $\mathbb{R}^2$.

EXERCISE 7.2

Let $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times m}$ satisfy $AB = I_m$. Please find the ranks of A and B.
What can be concluded about m and n?

WORKED SOLUTION

The rank inequalities give $\text{rank}(AB) = \text{rank}(I_m) = m \leq \text{rank}(A) \leq m$.
Thus $\text{rank}(A) = m$. For B, $m = \text{rank}(AB) \leq \text{rank}(B) \leq m$, so $\text{rank}(B) = m$ as well.
Since $\text{rank}(A) = m \leq \min(m, n)$, we obtain $m \leq n$.
The matrices are not assumed square, so one cannot conclude that B is an inverse of A.

7.1.1 Reducing spanning sets and extending independent sets to bases in $\mathbb{F}^n$

Every spanning set contains a basis, and every linearly independent set extends to a basis.

Reducing a spanning set to a basis

MOTIVATION AND GUIDING IDEA

1. **Method A:** We want a basis consisting of vectors from the original set (as in the preceding exercise).
   - Form a matrix whose **columns** are the vectors of S and reduce it to row echelon form.
   - For each pivot column, choose the corresponding **original column** as a basis vector. These columns are linearly independent.
   - The column space is not preserved by row operations; therefore the columns must be taken from the original matrix.
2. **Method B:** Find a **new** linearly independent set spanning the same space.
   - Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
   - Its nonzero echelon rows form a basis.

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- Row operations preserve the row space, so the basis may be taken after reduction.

Extending a linearly independent set to a basis

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- Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
- For each nonpivot column j, adjoin the standard basis vector e_j to S. To justify this construction, first replace the original rows by the echelon rows, preserving their span. At the pivot indices, the added standard basis vectors are zero, whereas the echelon rows give a triangular matrix with nonzero diagonal. Hence a vanishing linear combination first forces the coefficients of the echelon rows to be zero, and then those of the added standard basis vectors. Since the original rows are independent, replacing the echelon rows back by the originals preserves the basis property.

EXERCISE **7.3**

Let $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 2 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} \subseteq \mathbb{R}^5$

1. Please show that S is linearly independent.
2. Extend S to a basis of $\mathbb{R}^5$.

WORKED SOLUTION

1. Form a matrix with these vectors as rows, and row reduce:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{\text{row operations}} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}$$

   The echelon form has no zero row, so S is linearly independent.

2. Two more vectors are needed for a basis of $\mathbb{R}^5$. Form a matrix with the given vectors as **rows**, and reduce:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{\text{row operations}} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}$$

   The nonpivot columns are 2 and 4. Therefore adjoin to the **original** set S the corre-

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EXERCISE 7.3 CONTINUED

sponding standard basis vectors: e_2, e_4

$$B = S \cup \{e_2, e_4\} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 2 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

EXERCISE 7.4

1. Please find a basis B of $V = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$ over $\mathbb{R}$. What is its dimension?
2. Extend your basis B to a basis D of $\mathbb{R}^4$.

WORKED SOLUTION

1. Form a matrix whose rows are the given vectors, and row reduce:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{\text{row operations}} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since the vectors were entered as rows, row operations preserve their span. We may therefore choose the **nonzero rows** of the **echelon matrix**; they are linearly independent.

Thus the required basis is $B = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ -2 \\ -3 \end{pmatrix} \right\}$. It contains two vectors, so the dimension is 2.

2. Two further vectors are needed to obtain a basis of $\mathbb{R}^4$.

Write the vectors of B as rows. This matrix is already in echelon form, with pivot columns $1, 2$. Adjoin the standard basis vectors at the nonpivot indices $3, 4$, namely e_3, e_4:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The enlarged matrix has four independent rows, so the required basis is

$$D = B \cup \{e_3, e_4\} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ -2 \\ -3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

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EXERCISE 7.5 EXAM QUESTION

Work over $\mathbb{Z}_3 = \{0, 1, 2\}$, with all arithmetic modulo 3. For each $\omega \in \mathbb{Z}_3$, define

$$A(\omega) = \begin{pmatrix} 1 & \omega & 1 \\ \omega & 1 & 1 \\ 1 & 1 & \omega \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \qquad c = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

1. Please determine the values of ω for which $b \in \mathrm{Col}(A(\omega))$, and please find all solutions of $A(\omega)x = b$ in those cases.
2. Please determine the values of ω for which both $b \in \mathrm{Col}(A(\omega))$ and $c \in \mathrm{Col}(A(\omega))$ hold.

WORKED SOLUTION

A vector belongs to a matrix's column space if and only if the corresponding linear system is solvable. There are only three possible parameter values, so examine all three.

1. For $\omega = 0$, the system is

$$x_1 + x_3 = 1, \qquad x_2 + x_3 = 2, \qquad x_1 + x_2 = 0.$$

The first two equations give $x_1 = 1 - x_3$ and $x_2 = 2 - x_3$. Substitution into the third yields

$$(1 - x_3) + (2 - x_3) = 0 \iff -2x_3 = 0 \iff x_3 = 0,$$

because $3 = 0$ and $-2 = 1$ in $\mathbb{Z}_3$. The unique solution is therefore

$$x = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

For $\omega = 1$, all rows agree, so every vector in the column space has the form $(a, a, a)^T$. Since $b = (1, 2, 0)^T$ does not, there is no solution.

For $\omega = 2$, reduce the augmented matrix using arithmetic modulo 3:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 2 \end{array}\right).$$

The second row gives $2x_3 = 0$, hence $x_3 = 0$. The third gives $2x_2 = 2$, hence $x_2 = 1$. Finally, $x_1 + 2 = 1$, so $x_1 = 2$. The unique solution is

$$x = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

Thus $b \in \mathrm{Col}(A(\omega))$ exactly when $\omega \in \{0, 2\}$.

2. Expansion along the first row gives

$$\begin{aligned} \det A(\omega) &= (\omega - 1) - \omega(\omega^2 - 1) + (\omega - 1) \\ &= -\omega^3 + 3\omega - 2. \end{aligned}$$

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EXERCISE 7.5 EXAM QUESTION CONTINUED

Consequently $\det A(0) = 1$ and $\det A(2) = 2$ in $\mathbb{Z}_3$. Both matrices are invertible, so their column spaces are all of $\mathbb{Z}_3^3$. In particular, both b and c lie in them. For a direct check,

$$A(0)\begin{pmatrix}1\\0\\1\end{pmatrix} = c, \qquad A(2)\begin{pmatrix}0\\1\\0\end{pmatrix} = c.$$

For $\omega = 1$, we already know that b does not belong to the column space. The answer is therefore exactly $\omega \in \{0, 2\}$.

7.2 The null space and nullity of a matrix

DEFINITION 7.2

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m\times n}$.
The **null space of** A, also called its *kernel*, is the set of column vectors

$$N(A) := \{x \in \mathbb{F}^n \mid Ax = 0\},$$

that solve the homogeneous system $Ax = 0$.

- $N(A)$ is a vector subspace of $\mathbb{F}^n$.
- Its dimension is the **nullity** of A, denoted

$$\dim N(A) = \text{null}(A).$$

PROPOSITION 7.3

For $A \in \mathbb{F}^{m\times n}$,

$$\text{rank}(A) + \text{null}(A) = n.$$

- A has n columns, each a vector in $\mathbb{F}^m$.
- Reduce A to reduced row echelon form.
- The number of pivot columns equals $\text{rank}(A)$. The columns of the *original matrix* at these pivot indices form a basis of $\text{Col}(A)$; row reduction itself need not preserve the column space.
- There are $n - \text{rank}(A)$ nonpivot columns. These correspond to free variables in $Ax = 0$, each contributing one independent parameter to its solution space. Consequently,

$$\dim N(A) = \text{null}(A) = n - \text{rank}(A).$$

- Therefore,

$$\text{rank}(A) + \text{null}(A) = \text{rank}(A) + \big(n - \text{rank}(A)\big) = n.$$

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EXERCISE 7.6

Please find a basis and the dimension of the kernel of

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Please describe all solutions of $Ax = 0$ and please verify the rank-nullity theorem.

WORKED SOLUTION

The equations are $x_1 + 2x_2 = 0$, $x_2 + x_3 = 0$. Set $x_3 = t$; then

$$x_2 = -t, \qquad x_1 = 2t,$$

and hence

$$\ker A = \{t(2, -1, 1)^T : t \in \mathbb{R}\} = \text{span}\{(2, -1, 1)^T\}.$$

The spanning vector is nonzero, so it is a basis and the nullity is 1. The matrix has two pivot columns, so its rank is 2. Indeed, $\text{rank } A + \dim \ker A = 2 + 1 = 3$, the number of columns.

MOTIVATION AND GUIDING IDEA

Different questions require different spaces. The column space describes attainable right-hand sides. The null space describes changes in the input that leave the output unchanged. Rank and nullity measure these two kinds of freedom.

7.3 Interpreting matrix-vector multiplication

DEFINITION 7.3

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m \times n}$. Write its columns as

$$A = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix}, \qquad a_j \in \mathbb{F}^m.$$

Let $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{F}^n$.

1. **A linear combination of columns.** Matrix-vector multiplication is defined by

$$Av = v_1 a_1 + \cdots + v_n a_n.$$

Thus Av is the linear combination of the columns of A with coefficients $v_1, \dots, v_n$. It follows that

$$\{Av \mid v \in \mathbb{F}^n\} = \text{span}\{a_1, \dots, a_n\} = \text{Col}(A).$$

2. **A linear map.** The function

$$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(v) = Av,$$

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is linear: for all $u, v \in \mathbb{F}^n$ and $\lambda \in \mathbb{F}$,

$$T_A(u+v) = A(u+v) = Au + Av,$$

$$T_A(\lambda v) = A(\lambda v) = \lambda(Av).$$

Thus multiplication by A is a linear map, and its columns describe its action on the standard basis of $\mathbb{F}^n$:

$$T_A(e_j) = a_j.$$

EXERCISE 7.7

Let $\mathbb{F}$ be a field and $B \in \mathbb{F}^{n\times m}$. Define

$$J := \{Bv \mid v \in \mathbb{F}^m\} \subseteq \mathbb{F}^n.$$

Please show that

$$J = \mathrm{Col}(B).$$

WORKED SOLUTION

Denote the columns of B by

$$B = \begin{pmatrix} | & & | \\ b_1 & \dots & b_m \\ | & & | \end{pmatrix},$$

so that $b_1, \dots, b_m \in \mathbb{F}^n$ are its column vectors.
Now let

$$v = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} \in \mathbb{F}^m.$$

By the definition of matrix-vector multiplication,

$$Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m.$$

We prove both inclusions.

1. First, $J \subseteq \mathrm{Col}(B)$.

   Let $y \in J$. Then $y = Bv$ for some $v \in \mathbb{F}^m$.

   Every such vector is a linear combination of $b_1, \dots, b_m$:

   $$y = Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m$$

   for suitable scalars $\lambda_1, \dots, \lambda_m \in \mathbb{F}$.

   Thus y belongs to their span, namely

   $$y \in \mathrm{span}\{b_1, \dots, b_m\} = \mathrm{Col}(B).$$

   This proves $J \subseteq \mathrm{Col}(B)$.

2. Conversely, $\mathrm{Col}(B) \subseteq J$.

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EXERCISE 7.7 CONTINUED

Let $w \in \mathrm{Col}(B)$. There are scalars $\eta_1, \dots, \eta_m \in \mathbb{F}$ such that

$$w = \eta_1 b_1 + \cdots + \eta_m b_m.$$

Define

$$v := \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_m \end{pmatrix} \in \mathbb{F}^m.$$

The preceding calculation gives

$$Bv = \eta_1 b_1 + \cdots + \eta_m b_m = w.$$

Hence $w = Bv$ for some $v \in \mathbb{F}^m$, so

$$w \in J.$$

This proves $\mathrm{Col}(B) \subseteq J$.

The two inclusions yield

$$J = \mathrm{Col}(B),$$

as required.

EXERCISE 7.8

Let $\mathbb{F}$ be a field and let $A, B \in \mathbb{F}^{n \times n}$ satisfy

$$AB = 0.$$

Please show that

$$\mathrm{Col}(B) \subseteq N(A),$$

Every column of B, and every linear combination of its columns, is therefore annihilated by A.

WORKED SOLUTION

Use the definitions of column space and null space.
It suffices to prove that $Av = 0$ for every $v \in \mathrm{Col}(B)$.
By the definition of column space, $v \in \mathrm{Col}(B)$ implies that some $w \in \mathbb{F}^n$ satisfies

$$v = Bw.$$

Compute

$$Av = A(Bw) = (AB)w.$$

Since $AB = 0$,

$$(AB)w = 0w = 0,$$

and therefore

$$Av = 0.$$

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EXERCISE 7.8 CONTINUED

We have proved that $Av = 0$ for every $v \in \mathrm{Col}(B)$, or equivalently,

$$v \in N(A).$$

Consequently,

$$\mathrm{Col}(B) \subseteq N(A),$$

as required.

7.4 Exercises

EXERCISE 7.9

Let V be a vector space over a field $\mathbb{F}$, and let $S, R \subseteq V$ be arbitrary subsets, not necessarily finite.
Please write

$$\mathrm{span}(S) = \left\{ \sum_{i=1}^{k} \alpha_i s_i \;\middle|\; k \in \mathbb{Z}_{\geq 0},\ \alpha_i \in \mathbb{F},\ s_i \in S \right\},$$

The empty sum is the zero vector, so $\mathrm{span}(\varnothing) = \{0\}$. Define $\mathrm{span}(R)$ and $\mathrm{span}(S \cup R)$ similarly.
Please show that, for all $S, R \subseteq V$,

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R).$$

WORKED SOLUTION

Write

$$U := \mathrm{span}(S), \qquad W := \mathrm{span}(R).$$

Recall that the sum $U + W$ is defined by

$$U + W := \{u + w \mid u \in U,\ w \in W\}.$$

We prove

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R)$$

by establishing both inclusions:

$$\mathrm{span}(S) + \mathrm{span}(R) \subseteq \mathrm{span}(S \cup R) \quad \text{and} \quad \mathrm{span}(S \cup R) \subseteq \mathrm{span}(S) + \mathrm{span}(R).$$

1. First prove

$$\mathrm{span}(S) + \mathrm{span}(R) \subseteq \mathrm{span}(S \cup R).$$

Let $v \in \mathrm{span}(S) + \mathrm{span}(R)$. Then $v = u + w$ for some $u \in \mathrm{span}(S)$ and $w \in \mathrm{span}(R)$:

$$v = u + w.$$

By the definition of $\mathrm{span}(S)$, there are vectors $s_1, \ldots, s_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$ such that

$$u = \sum_{i=1}^{k} \alpha_i s_i.$$

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EXERCISE 7.9 CONTINUED

Similarly, there are vectors $r_1, \ldots, r_\ell \in R$ and scalars $\beta_1, \ldots, \beta_\ell \in \mathbb{F}$ such that

$$w = \sum_{j=1}^{\ell} \beta_j r_j.$$

Therefore,

$$v = u + w = \sum_{i=1}^{k} \alpha_i s_i + \sum_{j=1}^{\ell} \beta_j r_j.$$

Every s_i and every r_j lies in $S \cup R$. Thus v is a finite linear combination of vectors in $S \cup R$.

By the definition of span,

$$v \in \text{span}(S \cup R).$$

Hence

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$

2. For the reverse inclusion, prove

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

Let $v \in \text{span}(S \cup R)$. There are $x_1, \ldots, x_t \in S \cup R$ and $\gamma_1, \ldots, \gamma_t \in \mathbb{F}$ such that

$$v = \sum_{i=1}^{t} \gamma_i x_i.$$

Partition the indices into disjoint sets: assign to S every vector belonging to S, and assign the rest to R. A vector in the intersection is counted only in the first set.

Define

$$I_S := \{i \in \{1, \ldots, t\} \mid x_i \in S\}, \qquad I_R := \{1, \ldots, t\} \setminus I_S.$$

Then v can be written as

$$v = \sum_{i \in I_S} \gamma_i x_i + \sum_{i \in I_R} \gamma_i x_i.$$

Set

$$u := \sum_{i \in I_S} \gamma_i x_i, \qquad w := \sum_{i \in I_R} \gamma_i x_i.$$

Observe that

- for $i \in I_S$, $x_i \in S$, so u is a linear combination of vectors in S; thus

$$u \in \text{span}(S).$$

- for $i \in I_R$, $x_i \in R$, so w is a linear combination of vectors in R; thus

$$w \in \text{span}(R).$$

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EXERCISE 7.9 CONTINUED

We have obtained

$$v = u + w$$

with $u \in \text{span}(S)$ and $w \in \text{span}(R)$, hence

$$v \in \text{span}(S) + \text{span}(R).$$

This proves

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

The two inclusions give

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R),$$

as required.

EXERCISE 7.10

Let $\mathbb{R}_3[x]$ denote the real vector space of polynomials of degree at most 3.
Define the following subspaces of $\mathbb{R}_3[x]$:

$$U = \text{span}\{1,\ x,\ 1+x\},$$

$$W = \{\, p(x) \in \mathbb{R}_3[x] \mid p(0) = 0,\ p(1) = 0 \,\}.$$

Please find bases and dimensions of $U + W$ and $U \cap W$.

WORKED SOLUTION

We work in $\mathbb{R}_3[x]$.

1. First consider U.

   We have

   $$U = \text{span}\{1, x, 1+x\}.$$

   The linear relation

   $$(1+x) - 1 - x = 0,$$

   expresses $1+x$ as a linear combination of 1 and x. Thus $\{1, x, 1+x\}$ is linearly dependent.

   Conversely, $\{1, x\}$ is linearly independent: if

   $$a \cdot 1 + b \cdot x = 0$$

   as a polynomial identity, then comparison of coefficients gives

   $$a = 0, \quad b = 0.$$

   Thus $\{1, x\}$ is a basis of U, and

   $$\dim U = 2.$$

2. Now consider W.

   By definition,

   $$W = \{p(x) \in \mathbb{R}_3[x] \mid p(0) = 0,\ p(1) = 0\}.$$

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EXERCISE 7.10 CONTINUED

Write a general polynomial in $\mathbb{R}_3[x]$ as

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

The condition $p(0) = 0$ gives

$$p(0) = a_0 = 0,$$

and hence

$$p(x) = a_1x + a_2x^2 + a_3x^3.$$

The condition $p(1) = 0$ gives

$$p(1) = a_1 + a_2 + a_3 = 0.$$

Introduce parameters by setting

$$a_2 = s, \quad a_3 = t,$$

so that

$$a_1 = -s - t.$$

Every polynomial in W therefore has the form

$$p(x) = (-s - t)x + sx^2 + tx^3.$$

Equivalently,

$$p(x) = s(x^2 - x) + t(x^3 - x).$$

Hence

$$W = \text{span}\{x^2 - x,\ x^3 - x\}.$$

To show that this is a basis, suppose

$$\alpha(x^2 - x) + \beta(x^3 - x) = 0$$

as a polynomial identity. Collecting coefficients gives

$$\beta x^3 + \alpha x^2 + (-\alpha - \beta)x = 0.$$

A polynomial is zero if and only if all of its coefficients are zero:

$$\beta = 0, \quad \alpha = 0, \quad -\alpha - \beta = 0.$$

Thus $\alpha = \beta = 0$, so $x^2 - x$ and $x^3 - x$ are linearly independent.
Therefore,

$$\{x^2 - x,\ x^3 - x\}$$

is a basis of W, and

$$\dim W = 2.$$

3. Next find a basis and the dimension of $U + W$.

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EXERCISE 7.10 CONTINUED

The calculations above show that

$$U = \text{span}\{1, x\}, \quad W = \text{span}\{x^2 - x,\ x^3 - x\},$$

and hence

$$U + W = \text{span}\{1,\ x,\ x^2 - x,\ x^3 - x\}.$$

Set

$$S = \{1,\ x,\ x^2 - x,\ x^3 - x\}.$$

To test whether S is linearly independent, suppose

$$c_1 \cdot 1 + c_2 x + c_3(x^2 - x) + c_4(x^3 - x) = 0$$

as a polynomial identity. Collecting coefficients,

$$c_1 + (c_2 - c_3 - c_4)x + c_3 x^2 + c_4 x^3 = 0.$$

all coefficients must vanish:

$$\begin{cases} c_1 = 0, \\ c_2 - c_3 - c_4 = 0, \\ c_3 = 0, \\ c_4 = 0. \end{cases}$$

The third and fourth equations give $c_3 = c_4 = 0$, then the second gives $c_2 = 0$ and the first gives $c_1 = 0$. Only the trivial combination vanishes, so S is linearly independent.

There are four polynomials in S and $\dim \mathbb{R}_3[x] = 4$. Thus S is a basis of $\mathbb{R}_3[x]$, and

$$U + W = \mathbb{R}_3[x], \quad \dim(U + W) = 4.$$

4. Finally find a basis and the dimension of $U \cap W$.

   The dimension formula for a sum gives

   $$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

   Substituting,

   $$4 = 2 + 2 - \dim(U \cap W),$$

   Therefore,

   $$\dim(U \cap W) = 0.$$

   For a direct check, a polynomial in U has the form

   $$p(x) = a + bx,$$

   because $U = \text{span}\{1, x\}$.

   To belong to W as well, it must satisfy

   $$p(0) = 0, \quad p(1) = 0.$$

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EXERCISE 7.10 CONTINUED

From $p(0) = 0$,

$$a = 0.$$

and from $p(1) = 0$,

$$a + b = 0.$$

Substituting $a = 0$ into the second equation gives $b = 0$. The only polynomial in the intersection is therefore

$$p(x) \equiv 0.$$

Hence

$$U \cap W = \{0\}, \quad \dim(U \cap W) = 0.$$

The empty set is a basis of this intersection, since it contains no nonzero vector.

To summarize,

$$\dim(U + W) = 4, \quad U + W = \mathbb{R}_3[x],$$
$$\dim(U \cap W) = 0, \quad U \cap W = \{0\}.$$

EXERCISE 7.11

An engineering system seeks to recover three real state variables x_1, x_2, x_3 from four sensor readings. They may represent temperature deviations, loads or electrical signals in normalized units. Assume an exact, noise-free linear model:

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix}, \quad \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = M \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

1. Please find bases of the row and column spaces of M, and their dimensions.
2. How many readings are linearly dependent on the others?
3. Can all three state variables x_1, x_2, x_3 be recovered uniquely from the readings?
4. Suppose the readings are

$$\mathbf{y} = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 8 \end{pmatrix}.$$

First test whether $M\mathbf{x} = \mathbf{y}$ is consistent. Then replace only the fourth reading by 7 and please find all states compatible with the corrected readings.

WORKED SOLUTION

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EXERCISE 7.11 CONTINUED

1. Row reduce M:

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus its rank is

$$\operatorname{rank} M = 2.$$

The first two original rows are independent, and the third and fourth are linear combinations of them:

$$(1,1,2) = (1,0,1) + (0,1,1),$$

$$(2,1,3) = (1,0,1) + (1,1,2) = 2(1,0,1) + (0,1,1).$$

A basis of the row space is therefore

$$\{(1,0,1),(0,1,1)\}, \quad \dim(\text{Row}(M)) = 2.$$

The columns are

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 3 \end{pmatrix}.$$

Observe that

$$\mathbf{c}_3 = \mathbf{c}_1 + \mathbf{c}_2.$$

Since $\mathbf{c}_1, \mathbf{c}_2$ are independent,

$$\{\mathbf{c}_1, \mathbf{c}_2\}$$

is a basis of the column space. Its dimension is also 2, in agreement with equality of row rank and column rank.

2. The third and fourth rows are combinations of the first two, so the last two readings are linearly redundant. There are only two independent sensor readings.

3. The matrix is 4×3 and has rank 2. There are three unknowns, so

$$\dim(\ker M) = 3 - \operatorname{rank} M = 3 - 2 = 1.$$

The homogeneous solution space has dimension 1 and contains nonzero vectors. For each attainable $\mathbf{y}$, the solutions of $M\mathbf{x} = \mathbf{y}$ form an affine line, rather than a single point. Consequently the readings do not determine the state uniquely.

4. First test consistency. The row relations imply that every attainable reading vector satisfies

$$y_3 = y_1 + y_2, \qquad y_4 = 2y_1 + y_2.$$

For the original data $(2,3,5,8)^T$, the first relation holds, but the second requires $y_4 = 2 \cdot 2 + 3 = 7 \neq 8$. Thus the original system has **no solution**.

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EXERCISE 7.11 CONTINUED

Replacing the fourth reading by 7 gives the consistent system

$$M\mathbf{x} = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix}, \qquad \begin{cases} x_1 + x_3 = 2, \\ x_2 + x_3 = 3, \\ x_1 + x_2 + 2x_3 = 5, \\ 2x_1 + x_2 + 3x_3 = 7. \end{cases}$$

The last two equations now follow from the first two, including their right-hand sides. Set $t = x_3$ to obtain

$$x_1 = 2 - t, \qquad x_2 = 3 - t, \qquad x_3 = t, \qquad t \in \mathbb{R}.$$

The solution set is the affine line

$$\left\{ \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \,\middle|\, t \in \mathbb{R} \right\}.$$

Its direction vector $(-1, -1, 1)^T$ spans $N(M)$, as predicted by rank-nullity.

EXERCISE 7.12

In a calibration system, three measured signals (m_1, m_2, m_3) depend on three state variables (q_1, q_2, q_3). This abstract model applies whenever measurements are cumulative sums of the required quantities. Their linear relationship is

$$\begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix} = C \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

1. Please find a basis of the column space of C and please show that its columns form a basis of $\mathbb{R}^3$.
2. Please prove that C is invertible.
3. Please find C^{-1} explicitly by solving $C\mathbf{q} = \mathbf{m}$ for $\mathbf{q}$ in terms of $\mathbf{m}$.
4. Please express (q_1, q_2, q_3) in terms of (m_1, m_2, m_3).

WORKED SOLUTION

1. The columns of C are

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

To test linear independence, suppose

$$a\mathbf{c}_1 + b\mathbf{c}_2 + c\mathbf{c}_3 = \mathbf{0}.$$

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EXERCISE 7.12 CONTINUED

Equivalently,

$$a\begin{pmatrix}1\\1\\1\end{pmatrix}+b\begin{pmatrix}0\\1\\1\end{pmatrix}+c\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix},$$

giving the system

$$\begin{cases}a=0,\\a+b=0,\\a+b+c=0.\end{cases}$$

The first equation gives $a=0$, the second gives $b=0$ and the third gives $c=0$. Thus the columns are independent.

Three independent vectors in $\mathbb{R}^3$ form a basis, so

$$\text{Col}(C)=\mathbb{R}^3,\quad \text{rank}\,C=3.$$

2. C is a 3×3 matrix, and the preceding part gives

$$\text{rank}\,C=3.$$

Thus C has full rank and is invertible. In terms of linear systems, the homogeneous system $C\mathbf{q}=\mathbf{0}$ has

$$\dim(\ker C)=3-\text{rank}\,C=3-3=0.$$

so its unique solution is $\mathbf{q}=\mathbf{0}$.

Since $\text{Col}(C)=\mathbb{R}^3$, the system $C\mathbf{q}=\mathbf{m}$ is solvable for every $\mathbf{m}\in\mathbb{R}^3$.

Thus there is a *unique solution* $\mathbf{q}$ for every $\mathbf{m}$. Multiplication by C is a bijection from $\mathbb{R}^3$ to itself, and there is a matrix C^{-1} satisfying

$$C^{-1}C=CC^{-1}=I_3.$$

Therefore C is invertible. This relates matrix invertibility to injectivity and surjectivity of the corresponding linear map.

3. To compute C^{-1} explicitly, start with

$$\begin{pmatrix}1&0&0\\1&1&0\\1&1&1\end{pmatrix}\begin{pmatrix}q_1\\q_2\\q_3\end{pmatrix}=\begin{pmatrix}m_1\\m_2\\m_3\end{pmatrix}.$$

The resulting equations are

$$\begin{cases}q_1=m_1,\\q_1+q_2=m_2,\\q_1+q_2+q_3=m_3.\end{cases}$$

From the first equation,

$$q_1=m_1.$$

from the second,

$$m_1+q_2=m_2\quad\Longrightarrow\quad q_2=m_2-m_1.$$

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EXERCISE 7.12 CONTINUED

and from the third,

$$m_1 + (m_2 - m_1) + q_3 = m_3 \quad \Longrightarrow \quad m_2 + q_3 = m_3 \quad \Longrightarrow \quad q_3 = m_3 - m_2.$$

Therefore,

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}.$$

Hence

$$C^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

4. Since C is invertible, its columns form a basis of $\mathbb{R}^3$. Every measurement vector $\mathbf{m}$ is therefore a unique linear combination of those columns:

$$\mathbf{m} = C\mathbf{q}$$

with a unique $\mathbf{q}$. In this exact model, each measurement vector (m_1, m_2, m_3) determines a *unique state* (q_1, q_2, q_3), computed using C^{-1}:

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = C^{-1} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}.$$

This calibration matrix recovers the underlying quantities from their measurements.

EXERCISE 7.13

Let $\mathbb{F}$ be a field, and let $P_{\leq 3}$ be the space of polynomials over $\mathbb{F}$ of degree at most 3. Please write a polynomial in the standard basis as

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3,$$

with coefficient vector

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} \in \mathbb{F}^4.$$

Define three polynomial measurements:

$$\ell_1(p) = p(0), \quad \ell_2(p) = p(1), \quad \ell_3(p) = p'(0).$$

Represent them by a matrix $A \in \mathbb{F}^{3\times 4}$ such that, for every p,

$$A \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} \ell_1(p) \\ \ell_2(p) \\ \ell_3(p) \end{pmatrix}.$$

1. Please find A explicitly.

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EXERCISE 7.13 CONTINUED

2. Please find bases and dimensions of its row space and column space.
3. Please determine whether there are polynomials $p \in P_{\leq 3}$ satisfying
$$p(0) = 1, \quad p(1) = 1, \quad p'(0) = 0.$$
and, if so, please find all of them.
4. Please determine the affine dimension of the solution set. How many independent parameters remain?

WORKED SOLUTION

Write
$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3.$$

1. Compute the three measurements:

   Evaluation at 0 gives
   $$p(0) = a_0.$$
   and evaluation at 1 gives
   $$p(1) = a_0 + a_1 + a_2 + a_3.$$
   The formal derivative is
   $$p'(x) = a_1 + 2a_2 x + 3a_3 x^2,$$
   so
   $$p'(0) = a_1.$$
   Therefore,
   $$A \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_0 \\ a_0 + a_1 + a_2 + a_3 \\ a_1 \end{pmatrix}.$$
   and hence
   $$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$
2. Begin the row reduction of A:
   $$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$
   Subtract the first row from the second:
   $$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

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EXERCISE 7.13 CONTINUED

Subtract the third row from the second:

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Reorder the rows to obtain echelon form:

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

There are three nonzero echelon rows, so

$$\operatorname{rank}(A) = 3.$$

These rows span the row space. A convenient basis is

$$\{(1,0,0,0), (0,1,0,0), (0,0,1,1)\}.$$

The pivot indices are $1, 2, 3$. The corresponding columns of the original matrix therefore form a basis of its column space. The columns are

$$c_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad c_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad c_4 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Immediately,

$$c_4 = c_3.$$

Thus one may choose the column-space basis

$$\{c_1, c_2, c_3\},$$

whose dimension is

$$\dim(\operatorname{Col}(A)) = 3.$$

3. We seek p satisfying

$$p(0) = 1, \quad p(1) = 1, \quad p'(0) = 0.$$

In matrix form this is the system

$$A \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Explicitly,

$$\begin{cases} a_0 = 1, \\ a_0 + a_1 + a_2 + a_3 = 1, \\ a_1 = 0. \end{cases}$$

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EXERCISE 7.13 CONTINUED

The first equation gives

$$a_0 = 1.$$

the third gives

$$a_1 = 0.$$

and substitution into the second gives

$$1 + 0 + a_2 + a_3 = 1.$$

Hence

$$a_2 + a_3 = 0.$$

This is one equation in a_2, a_3, leaving one free parameter. An infinite field gives infinitely many solutions; a finite field $\mathbb{F}$ gives exactly $|\mathbb{F}|$ solutions. Set

$$t = a_2,$$

so that

$$a_3 = -t.$$

Therefore,

$$a_0 = 1, \quad a_1 = 0, \quad a_2 = t, \quad a_3 = -t,$$

and all the required polynomials have the form

$$p(x) = 1 + tx^2 - tx^3, \quad t \in \mathbb{F}.$$

4. There are four unknowns and the rank is 3. By rank-nullity,

$$\text{null}(A) = 4 - \text{rank}(A) = 4 - 3 = 1.$$

The null space has dimension 1. Whenever $Ax = b$ is consistent, its solution set is a translate of the kernel, an affine line of dimension 1; it need not be a vector subspace.

Here the system is consistent, and the free parameter t parametrizes that line. Thus the affine dimension of the polynomial solution set is 1, in agreement with rank-nullity.

EXERCISE 7.14

Two sensors measure Ax for $x \in \mathbb{R}^3$, where

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

An additional sensor measures $z = ax_1 + bx_2 + cx_3$. Please find a necessary and sufficient condition for unique recovery, and please give one useful sensor and one whose reading follows from the existing readings.

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EXERCISE 7.14 CONTINUED

WORKED SOLUTION

First find the changes that are invisible to the two original sensors:

$$Av = 0 \iff v_1 + 2v_3 = 0,\ v_2 - v_3 = 0 \iff v = t(-2, 1, 1)^T.$$

On such a change, the new sensor returns

$$a(-2t) + bt + ct = (c - 2a + b)t.$$

The three readings determine the state uniquely exactly when this expression vanishes only for $t = 0$, namely when

$$c - 2a + b \neq 0.$$

The determinant gives the same condition: $\det \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ a & b & c \end{pmatrix} = c + b - 2a$. The sensor $z = x_3$ is useful. In contrast, $z = x_1 + x_2 + x_3$ is the sum of the two existing readings and adds no information; for $(a, b, c) = (1, 1, 1)$, the condition indeed fails.

EXERCISE 7.15 EXAM QUESTION

1. Let $n \geq 2$ and $A, B \in M_n(\mathbb{R})$ satisfy $B \neq 0$ and $\operatorname{rank}(A) = n - 1$. Please prove that $AB = 0$ implies $\operatorname{rank}(B) = 1$.
2. Let $A, B \in M_{2026}(\mathbb{R})$ satisfy $AB = 0$. Please prove that
$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq 2026.$$

WORKED SOLUTION

The key implication in both parts is

$$AB = 0 \quad \Longrightarrow \quad \operatorname{Im}(B) \subseteq \ker A.$$

Indeed, every $y \in \operatorname{Im} B$ has the form $y = Bx$. Thus $Ay = A(Bx) = (AB)x = 0$, so $y \in \ker A$.

1. The inclusion and rank-nullity give
$$\begin{aligned} \operatorname{rank}(B) &= \dim \operatorname{Im}(B) \\ &\leq \dim \ker A \\ &= n - \operatorname{rank}(A) \\ &= n - (n - 1) = 1. \end{aligned}$$
Since $B \neq 0$, at least one column of B is nonzero, so $\operatorname{rank}(B) \geq 1$. The two bounds yield $\operatorname{rank}(B) = 1$.
2. Apply the same inclusion in $\mathbb{R}^{2026}$:
$$\begin{aligned} \operatorname{rank}(B) &= \dim \operatorname{Im}(B) \\ &\leq \dim \ker A \\ &= 2026 - \operatorname{rank}(A). \end{aligned}$$
Rearranging gives
$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq 2026.$$

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EXERCISE 7.15 EXAM QUESTION CONTINUED

If every output of B is annihilated by A, the entire image of B must lie in the kernel of A. The dimension of that kernel therefore bounds the rank of B.

Additional practice

EXERCISE 7.16 FOUNDATION

Over $\mathbb{R}$, consider

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 2 & 1 & 0 \end{pmatrix}.$$

Please find bases of its row space, column space and kernel, and please verify rank-nullity. Please give a condition on $b = (b_1, b_2, b_3)^T$ for $Ax = b$ to be solvable, and please find all solutions for $b = (3, 2, 5)^T$.

WORKED SOLUTION

The third row is the sum of the first two, which are independent. Thus a row-space basis is

$$\{(1, 2, 0, 1), (0, 0, 1, -1)\}, \qquad \text{rank } A = 2.$$

Original columns $1, 3$ are independent and form a column-space basis:

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

In the homogeneous system, set $x_2 = s, x_4 = t$. Then $x_1 = -2s - t$ and $x_3 = t$, so

$$\ker A = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

These vectors are independent by their second and fourth entries. The nullity is 2, and $2+2 = 4$. The row relation requires $b_3 = b_1 + b_2$. This is also sufficient, since the first two equations are solvable for all b_1, b_2. For the given right-hand side, all solutions are

$$x = \begin{pmatrix} 3 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \qquad s, t \in \mathbb{R}.$$

This is an affine plane, not a vector subspace, since $b \neq 0$.

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EXERCISE 7.17 CHALLENGE

Let $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times p}$. Please prove

$$\operatorname{rank}(AB) = \operatorname{rank}(B) - \dim\big(\operatorname{Col}(B) \cap \ker A\big),$$

and please deduce $\operatorname{rank}(AB) \geq \operatorname{rank}(A) + \operatorname{rank}(B) - n$. Please verify the identity for

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{pmatrix}$$

over $\mathbb{R}$.

WORKED SOLUTION

Restrict multiplication by A to the linear map $L : \operatorname{Col}(B) \to \mathbb{F}^m$, $L(y) = Ay$. Its domain has dimension $\operatorname{rank}(B)$, its image is $\operatorname{Col}(AB)$, and

$$\ker L = \operatorname{Col}(B) \cap \ker A.$$

Rank-nullity for L gives the identity. Since the intersection has dimension at most $\dim \ker A = n - \operatorname{rank}(A)$, the required inequality follows. In the example, the columns of B are independent, so its rank is 2. Moreover,

$$\ker A = \operatorname{span}\{(-1,-1,1)^T\} \subseteq \operatorname{Col}(B), \qquad AB = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}.$$

The inclusion follows from $(-1,-1,1)^T = -Be_1 - Be_2$. The intersection has dimension 1, giving $\operatorname{rank}(AB) = 2 - 1 = 1$, as the product confirms. Exactly those output directions of B that lie in the kernel of A are lost.

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EXERCISE 7.18 CHALLENGE

Let $A \in M_n(F)$ over any field. Please prove

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) \geq n,$$

with equality if and only if $A^2 = A$.

WORKED SOLUTION

Put $U = \operatorname{Col} A$ and $W = \operatorname{Col}(I_n - A)$. Since $x = Ax + (I_n - A)x$, we have $U + W = F^n$. The dimension formula gives

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) = n + \dim(U \cap W) \geq n.$$

If equality holds, then $U \cap W = \{0\}$. For every x, the vector $A(I_n - A)x = (I_n - A)Ax$ belongs to both spaces, so it is zero. Hence $A^2 = A$. Conversely, assume $A^2 = A$ and let $y = Au = (I_n - A)v$. The first representation gives $Ay = y$, while the second gives $Ay = (A - A^2)v = 0$. Therefore $y = 0$, the intersection is trivial, and equality follows.

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EXERCISE 7.19 CHALLENGE

Let $m, n \geq 2$ and let $W \neq \{0\}$ be a subspace of $M_{m\times n}(F)$ in which every matrix has rank at most 1. Please prove that at least one of the following holds:

1. There is a fixed $u \in F^m \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $v \in F^n$.
2. There is a fixed $v \in F^n \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $u \in F^m$.

WORKED SOLUTION

Every nonzero rank-one matrix is uv^T with $u, v \neq 0$: choose a nonzero column u and collect the scalar multiples expressing the other columns into v^T. First consider uv^T and rs^T. If both u, r and v, s are independent pairs, the matrix with rows v^T, s^T has rank 2 and maps onto F^2. Thus there are x, y with

$$v^T x = 1,\ s^T x = 0, \qquad v^T y = 0,\ s^T y = 1.$$

The sum sends x to u and y to r, so its rank is at least 2. Consequently a sum of rank at most 1 forces dependence of at least one pair of factors. Choose $A_0 = u_0 v_0^T \in W$ nonzero. If every column space lies in span$\{u_0\}$, the first alternative holds. Otherwise choose B whose column direction is independent of u_0. Since $A_0 + B \in W$, the preceding argument forces the row factor of B to be a multiple of v_0. Absorbing that nonzero scalar into its column factor gives $B = rv_0^T$ with r, u_0 independent. For any nonzero $C = cd^T \in W$, suppose d, v_0 were independent. Applying the same argument to $A_0 + C$ forces $c \in$ span$\{u_0\}$; applying it to $B + C$ forces $c \in$ span$\{r\}$. This gives $c = 0$, a contradiction. Every row factor is therefore a multiple of v_0, proving the second alternative.

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